Temporal Regularized Learning: Deep Slow-Feature Learning Local in Space and Time

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Abstract

Slow Feature Analysis (SFA) learns persistent factors by making adjacent representations similar while preserving their variance and decorrelating their coordinates. Can these three requirements train a deep nonlinear encoder without carrying an error through depth or time? We derive a regularized target from a soft SFA objective and map its residual to the preactivation of each neuron. Without lateral coupling, the outer product of this neuron state and presynaptic activity is the exact feedforward gradient. One pass through a learned inhibitory operator modifies the state used for the feedforward update, and pairs of the same states drive an anti-Hebbian lateral rule.

The resulting method, Temporal Regularized Learning (TeReL), processes one observation at a time, retains fixed-size detached state, and sends no error between layers. Its encoder uses plain stochastic gradient descent without optimizer state. On label-ordered MNIST, it reaches $95.84 \pm 0.07\%$ held-out accuracy after two data presentations. On a train-derived validation split, residual-state inhibition improves a matched samplewise reference by 1.42 percentage points while preserving effective rank. A minibatch relaxation, TeReL-Offline, reaches $97.30 \pm 0.07\%$, compared with $98.34 \pm 0.08\%$ for presentation-matched backpropagation and $96.98 \pm 0.10\%$ for Local SupCon. Objective ablations and a direct-covariance control in the minibatch relaxation identify the roles of the three regularizers and the lagged decorrelation signal. The evidence establishes local learning under a controlled temporal relation; it does not show that arbitrary natural adjacency is informative.

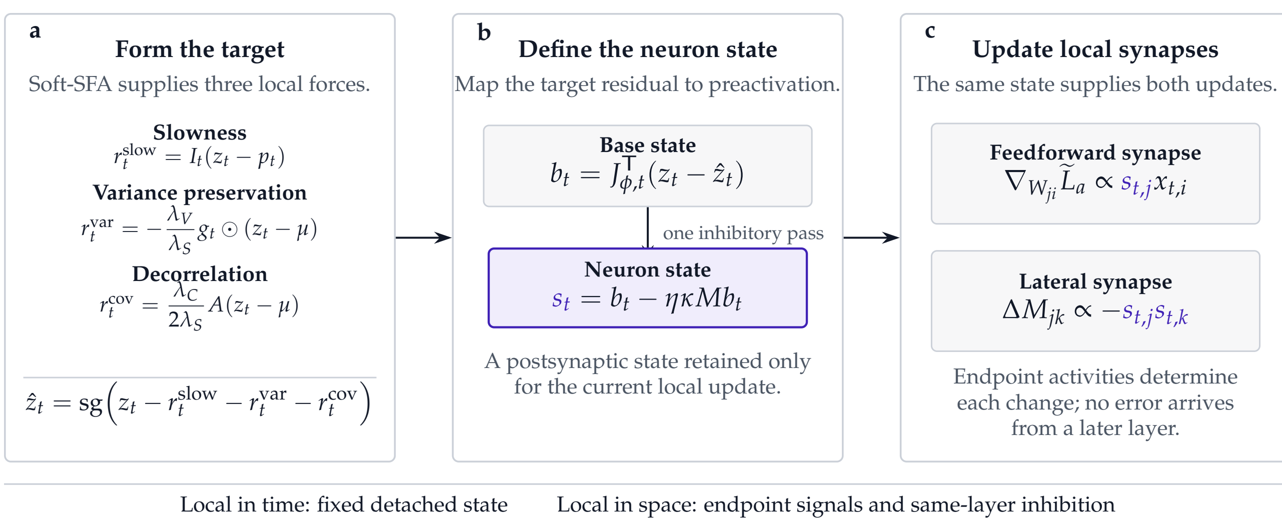

Figure 1: TeReL converts soft-SFA forces into neuron-local synaptic signals. **a**, The forces define a detached target. **b**, Its preactivation residual becomes the neuron state, followed by one inhibitory pass. **c**, Presynaptic activity and neuron state form the feedforward gradient; pairs of neuron states drive the anti-Hebbian lateral change.

1 Introduction

The observations in a temporal stream may change faster than the causes that produce them. A useful representation should then move slowly. Slowness alone, however, has an immediate failure: every observation can map to the same point. SFA removes this solution by requiring nonzero variance and decorrelated coordinates Wiskott and Sejnowski [2002]. These requirements define the learning problem: slow, noncollapsed, and nonredundant representations.

Our question is whether they also define a learning rule that is local in space and time. Spatial locality means that a synapse receives quantities available at its endpoints, possibly after same-layer neural dynamics, but no error from a later layer. Temporal locality means that the update reads a fixed detached state rather than an activation graph extending into the past. These are restrictions on credit assignment, not claims of neuron independence or biological realism.

TEREL realizes both restrictions by working backward from the soft-SFA gradient. The three objective terms first define a detached regularized target. Each neuron then tracks its preactivation residual to this target. For a linear synapse, the gradient factors into the neuron's residual state and the presynaptic activity. Their running pairwise second moment defines a dense inhibitory operator. Both factors of the feedforward rule are available at the synapse, giving it a Hebbian two-factor form; the signed lateral change is anti-Hebbian. Section 2 states both signs explicitly. Plain stochastic gradient descent applies this outer product directly; no optimizer moments alter the synaptic update.

Inhibitory correction of the residual state is the central distinction between TEREL and the exact detached soft-SFA gradient. In a matched batch-size-one experiment, it raises validation accuracy by 1.42 points without sacrificing effective rank. TEREL-Offline provides a less restrictive performance reference. It retains adjacent activations in a short computation graph and uses minibatch normalization; it is not the definition of the method.

The source of temporal order bounds what these experiments establish. We construct the MNIST stream from label-homogeneous chunks. The labels determine adjacency but do not enter the TEREL objective. This isolates a task-aligned temporal relation; it is not unlabeled self-supervision, and it supplies no evidence that arbitrary natural adjacency is informative.

The paper makes three contributions:

1. We derive a regularized target and a preactivation residual state whose outer product with presynaptic activity is the exact feedforward gradient of a detached surrogate.
2. We learn inhibitory lateral connections from the same neuron states and state the resulting locality boundary and $O(D^2)$ cost exactly.
3. We evaluate the samplewise rule, a batched relaxation, Local SupCon, direct covariance, and the objective mechanisms under one controlled MNIST construction, with selection separated from final evaluation.

2 Temporal Regularized Learning

We begin with one observation at one learned layer. Its soft-SFA objective defines a target; the residual to that target defines the neuron state used by the feedforward and lateral synapses.

2.1 A soft-SFA target

For observation t, let

$$a_t = Wx_t + c, \qquad z_t = \phi(a_t), \qquad z_t \in \mathbb{R}^D, \tag{1}$$

where x_t is detached from the preceding learned layer. Before processing x_t, the layer retains the preceding activation p_t, running mean μ, running variance σ^2, and off-diagonal second-moment operator A. All four are detached. The indicator I_t is zero at a sequence boundary and one otherwise. For this observation, define

$$\ell_S = \frac{I_t}{D}\|z_t - p_t\|_2^2 \qquad \ell_V = -\frac{1}{D}\sum_j [\gamma - \sigma_j^2]_+ (z_{t,j} - \mu_j)^2 \qquad \ell_C = \frac{1}{D}(z_t - \mu)^\top A \,\mathrm{sg}(z_t - \mu). \tag{2}$$

The layer minimizes $\ell = \lambda_S \ell_S + \lambda_V \ell_V + \lambda_C \ell_C$. The three terms favor temporal persistence, prevent constant coordinates, and oppose correlated coordinates, respectively. Stored quantities remain fixed during the update.

These three forces can be written as one target. Let $g = [\gamma - \sigma^2]_+$. From detached values at the current iterate, form

$$r_t = I_t(z_t - p_t) - \frac{\lambda_V}{\lambda_S} g \odot (z_t - \mu) + \frac{\lambda_C}{2\lambda_S} A(z_t - \mu), \tag{3}$$

$$\hat{z}_t = \mathrm{sg}(z_t - r_t). \tag{4}$$

The vector $\hat{z}_t$ is not an externally supplied label. It is the local activation that would cancel the instantaneous combination of temporal, variance, and decorrelation forces.

Proposition 1 (Target equivalence)**.** *At the iterate where Eq. (4) is constructed, the detached target loss*

$$\widetilde{\ell}_z = \frac{\lambda_S}{D}\|z_t - \hat{z}_t\|_2^2 \tag{5}$$

has the same derivative with respect to z_t as ℓ in Eq. (2).

The result follows by substituting $z_t - \hat{z}_t = r_t$ into $\partial \widetilde{\ell}_z / \partial z_t = 2\lambda_S r_t / D$. The target compresses the three local gradients without changing them.

2.2 From target residual to neuron state

The activation residual $r_t = z_t - \hat{z}_t$ is not yet a neuron state at the input to a feedforward synapse. We map it through the activation Jacobian,

$$b_t = J_{\phi,t}^\top r_t, \qquad J_{\phi,t} = \left.\frac{\partial \phi(a)}{\partial a}\right|_{a_t}. \tag{6}$$

For an elementwise activation, $b_t = r_t \odot \phi'(a_t)$. Thus $b_{t,j}$ is the preactivation residual tracked by neuron j, not an error delivered from another layer.

We next let same-layer inhibition act on this state. Let $M \succeq 0$ denote the running second moment of the neuron states, let $\kappa \geq 0$ set its coupling, and let $\eta > 0$ be the pass size. One lateral pass gives

$$s_t = b_t - \eta\kappa M b_t. \tag{7}$$

The correction requires one matrix–vector product and no iterative settling. Setting $\kappa = 0$ recovers $s_t = b_t$ and hence the exact soft-objective gradient. Nonzero inhibition deliberately modifies that gradient by subtracting the component $\eta\kappa M b_t$.

Now define the detached preactivation target $\hat{a}_t = \mathrm{sg}(a_t - s_t)$. Its mean-squared error gives

$$\widetilde{\ell}_a = \frac{\lambda_S}{D}\|a_t - \hat{a}_t\|_2^2, \qquad \frac{\partial \widetilde{\ell}_a}{\partial W} = \frac{2\lambda_S}{D} s_t x_t^\top, \qquad \frac{\partial \widetilde{\ell}_a}{\partial c} = \frac{2\lambda_S}{D} s_t. \tag{8}$$

The gradient has the intended two-factor Hebbian form: presynaptic activity times the state of the postsynaptic neuron. Plain gradient descent would use $\Delta W = -\alpha(2\lambda_S/D)s_t x_t^\top$, which is the update used by TEREL. We do not absorb the descent sign into the word “Hebbian.”

2.3 Anti-Hebbian learning on the same states

After the parameter update, the same detached neuron states update M:

$$M^{+} = \rho_M M + (1-\rho_M)\frac{1}{D} s_t s_t^{\top}. \tag{9}$$

The positive-semidefinite matrix M stores residual-state second moments and appears with a negative sign in Eq. (7). Equivalently, the signed lateral weight is $L^{\mathrm{lat}} = -\eta\kappa M$, whose update contains $-\eta\kappa(1-\rho_M)s_{t,j}s_{t,k}$. Positive coactivity therefore contributes a negative term to L^{lat}_{jk}, which is the anti-Hebbian convention used here. Both endpoints expose neuron states: the lateral update uses $s_{t,j}s_{t,k}$, not a residual paired with an unrelated activation. The diagonal of M is nonnegative and acts as learned self-inhibition; the off-diagonal entries connect distinct neurons and need not be elementwise nonnegative.

The representation operator A and the inhibitory operator M serve different purposes. The former constructs the decorrelation component of $\hat{z}_t$; the latter acts on the resulting neuron states. We retain A because replacing its representation statistic by residual-state covariance would change the objective rather than merely realize its gradient.

The operator A is a lagged proxy for the current representation covariance. For minibatch $\mathcal{B}_k$ of size B, using the mean μ_{k-1} available before that batch, let

$$C_k = \mathrm{offdiag}\left[\frac{1}{B}\sum_{t\in\mathcal{B}_k}(z_{k,t}-\mu_{k-1})(z_{k,t}-\mu_{k-1})^{\top}\right]. \tag{10}$$

The direct penalty $\|C_k\|_F^2/D$ has activation gradient $4C_k(z_{k,t}-\mu_{k-1})/(BD)$. Replacing C_k by stored A_{k-1} gives the direction in Eq. (3), up to a positive scale. The substitution is a lagged gradient proxy, not an equality between scalar objectives. The direct minibatch control in Section 5 tests whether the approximation matters in the offline relaxation.

2.4 Execution order and locality

The layer reads all state before it writes any state. After computing z_t, $\hat{z}_t$, and s_t, it updates W using Eq. (8). It then stores the current detached activation and updates

$$\mu^{+} = \rho_\mu \mu + (1-\rho_\mu) z_t, \tag{11}$$

$$(\sigma^2)^{+} = \rho_\mu \sigma^2 + (1-\rho_\mu)(z_t-\mu)^2, \tag{12}$$

$$A^{+} = \rho_A A + (1-\rho_A)\,\mathrm{offdiag}[(z_t-\mu)(z_t-\mu)^{\top}], \tag{13}$$

along with Eq. (9). A boundary flag invalidates the stored predecessor. No activation graph survives the step.

Require: observation x_t, boundary I_t, parameters W, c, detached state p, μ, σ^2, A, M
1: $a_t \leftarrow W\,\mathrm{sg}(x_t) + c$; $z_t \leftarrow \phi(a_t)$
2: construct $\hat{z}_t$ by Eqs. (3) and (4)
3: $b_t \leftarrow J_{\phi,t}^{\top}(z_t - \hat{z}_t)$
4: compute s_t with the lateral correction in Eq. (7)
5: update W, c from the detached target $\hat{a}_t \leftarrow \mathrm{sg}(a_t - s_t)$
6: update p, μ, σ^2, A, M from detached z_t, s_t

Algorithm 1: One samplewise TEREL update.

Algorithm 1 is applied independently at each learned layer in the same forward pass. A layer receives the detached activation of its predecessor and owns its target, neuron state, and running matrices.

At a feedforward synapse W_{ji}, Eq. (8) uses only the presynaptic activity x_i and postsynaptic state s_j. Computing s_j may use signals delivered by lateral connections M_{jk}. Each such connection updates from the states at its two endpoints. No error crosses a learned layer boundary, so the rule is neuron-local in space but not neuron-independent. The predecessor, moments, and lateral matrices are detached and fixed in size; the rule is also local in time.

2.5 The batched relaxation

TEREL-Offline is a performance reference that relaxes temporal locality. It optimizes Eq. (2) directly on class-homogeneous chunks, keeps $p_t = z_{t-1}$ live within a chunk, and uses BatchNorm. The temporal derivative then includes a future term. With $N_I = \sum_t I_t$ valid transitions, it is

$$\frac{2\lambda_S}{N_I D}\left[I_t(z_t - z_{t-1}) - I_{t+1}(z_{t+1} - z_t)\right]. \tag{14}$$

This derivative requires a short temporal graph. Training-mode BatchNorm also couples examples within each coordinate. BatchNorm is an optimization choice in this relaxation, not a fourth part of the method and not part of the TEREL locality claim.

Thus TEREL has a one-observation update and a fixed, detached temporal reference. TEREL-Offline, direct covariance, and Local SupCon instead couple examples within a chunk or minibatch; their error graphs are not temporally local in this sense.

2.6 State and compute cost

For a width-D layer, TEREL stores three causal scalars per neuron: p_j, μ_j, and σ_j^2, together with one layer-validity flag. The auxiliary operators $A, M \in \mathbb{R}^{D\times D}$ contribute $2D^2$ parameters; they are not temporal state. Target construction applies A once, and the inhibitory correction applies M once; updating the operators forms two rank-one outer products. In the reported encoder, the 655,360 auxiliary parameters are 1.23 times the 533,248 feedforward parameters. Optimizer moments and temporary activations form separate resource categories, and none of this training-only storage is needed by the encoder at inference.

3 Relation to prior work

SFA and incremental SFA. Temporal slowness, variance preservation, and decorrelation define the SFA objective Wiskott and Sejnowski [2002]. Batch SFA whitens observations and extracts minor components of their temporal differences. IncSFA replaces these batch operations by incremental PCA and minor-component updates, handles episodic boundaries, and admits Hebbian and anti-Hebbian forms Kompella et al. [2012]. Bio-SFA likewise derives an online neural system with local synaptic updates Lipshutz et al. [2020]. These methods establish that online and local SFA are possible. TEREL addresses a different question: how a soft-SFA target can train several nonlinear layers while blocking gradients at every learned layer and time boundary.

Deep gradient-based SFA also predates this work. PowerSFA differentiates through approximate whitening Schüler et al. [2018], while Slow and Steady Feature Analysis penalizes first- and higher-order changes across video tuples Jayaraman and Grauman [2016]. TEREL keeps only first-order adjacency and replaces global whitening by detached statistics and learned same-layer operators. Our classical baseline is a Python 3 port of the published IncSFA algorithm, checked against batch SFA on a synthetic stream.

Trace learning. Földiák-style trace learning exploits temporal persistence through a positive activation trace Földiák [1991]. That trace differs algebraically from the signed temporal residual in Eq. (3). The distinction matters here because this residual becomes both a postsynaptic feedforward factor and the signal at each end of a lateral synapse.

Gradient-isolated predictive and contrastive learning. Greedy InfoMax applies local InfoNCE objectives to gradient-isolated modules along the natural order of images, audio, and video Löwe et al. [2019]. CLAPP makes contrastive prediction causal and local across depth and time, using pre- and postsynaptic activity, predictive dendritic input, and broadcast modulation Illing et al. [2021]. LoCo instead permits the next block's loss to update an overlap between adjacent blocks Xiong et al. [2020]. TEREL uses no negative set or predictive decoder. It instead maintains two dense, lagged same-layer operators, and its main experiment obtains adjacency from labels rather than from an unlabeled stream.

Regularized and blockwise self-supervision. VICReg and Barlow Twins avoid collapse without negatives by regularizing variance and covariance or cross-correlation Bardes et al. [2022], Zbontar et al. [2021]. Such losses can train network blocks without end-to-end backpropagation Siddiqui et al. [2024]; layerwise VICReg has also been applied to pairs of original and augmented images with current-batch statistics Datta et al. [2025]. TEREL instead uses temporal pairs, detached running moments, and the target–state construction needed for one-observation execution.

The positive relation itself matters. Spectral analyses show that it can determine downstream geometry when it aligns with task labels Balestriero and LeCun [2022]. Our class-chunk MNIST experiment should be interpreted as constructed pair supervision and compared with a label-aware local baseline. It does not show that arbitrary natural adjacency reveals class structure.

The appendix compares the objectives, execution boundaries, and auxiliary storage of the nearest methods in a common table.

4 Experimental protocol

The experiments ask three questions. Does samplewise TEREL learn a useful, noncollapsed representation? Does the residual-state mechanism improve the same samplewise rule without inhibition? How does the batched relaxation compare with local and end-to-end baselines?

4.1 Data splits and roles

MNIST controlled temporal supervision. We stratify the 60,000-example MNIST training split into 50,000 examples for fitting and 10,000 for validation. Final accuracy is measured on the official held-out split. At each encoder epoch, training examples are randomly arranged into same-class chunks of length 16, with a boundary at every chunk start. Labels determine this order but never enter a TEREL objective. The protocol isolates a task-aligned temporal signal and is not unlabeled self-supervision.

Neuron state. We visualize neuron dynamics on CAPTURE-24, which records roughly 24 hours of wrist acceleration from each participant during daily life Chan et al. [2024]. Each consecutive 10-second interval is summarized by the mean, standard deviation, minimum, and maximum of the three axes and their magnitude. The trace comprises 128 windows from one uninterrupted fitting segment. We fixed the segment and randomly selected four neurons in the second layer before inspecting their values. The figure illustrates online dynamics, not improved recognition from natural order. Its archived data retain the exact interval, neuron indices, and state values.

4.2 Models, baselines, and readouts

All MNIST encoders have dimensions $784 \to 512 \to 256$ and use LeakyReLU with negative slope 0.01. TEREL trains both layers jointly, one sample at a time, for two data presentations. It uses identity normalization, plain stochastic gradient descent with learning rate 10^{-2} and no momentum or weight decay, $(\lambda_S, \lambda_V, \lambda_C) = (1, 2.5, 1)$, $\gamma = 1$, and momenta $\rho_\mu = \rho_A = \rho_M = 0.99$. Residual inhibition uses $\kappa = 1000$, one lateral pass of size 0.1, and the feature-normalized full second moment in Eq. (9). Every sample causes one optimizer update.

The matched samplewise reference omits the inhibitory pass, so $s_t = b_t$ and the update follows the exact detached soft-objective gradient. This comparison isolates the mechanism introduced in Section 2.2; architecture, order, normalization, optimizer, state timescales, and data presentations are identical.

TEREL-Offline uses the same hidden dimensions and activation, affine BatchNorm, live temporal pairs within each chunk, and greedy training for 30 epochs per layer, for 60 data presentations. AdamW uses learning rate 3×10^{-4}, weight decay 10^{-4}, and batch size 256. Its loss coefficients are $(1, 2.5, 1)$ and its population momentum is 0.9. The next minibatch uses the representation operator written by the preceding one. This configuration is an optimization-rich reference, not a spatially or temporally matched control for TEREL.

The backpropagation baseline matches the TEREL-Offline architecture, normalization, optimizer, batch size, and 60 data presentations, and trains end to end with cross entropy. Two random encoders separate learning from normalization. One omits normalization; the other keeps all hidden and affine BatchNorm parameters at initialization and updates only BatchNorm running statistics in one fitting-split pass. This calibration is data-dependent normalization, not representation learning.

Local SupCon is a label-aware local comparator for the constructed stream. Each layer minimizes supervised contrastive loss over all same-label positives in its minibatch, and activations are detached between layers. It matches the TEREL-Offline architecture, BatchNorm, batch size, 60 data presentations, all-layer probe, and evaluation runs. Its learning rate and temperature are selected from a fixed 3×2 validation grid.

Every primary representation is evaluated by the same linear classifier on the concatenation of both hidden layers. A second probe reads only the final layer of the identical TEREL-Offline encoder. All probes use AdamW for 60 epochs with learning rate 0.003, weight decay 10^{-4}, and batch size 2,048.

4.3 Model selection and evaluation separation

All selection uses the 10,000-example validation split. The declared samplewise screen varies optimizer structure, learning rate, and the number of lateral passes, with three validation runs for the final comparisons. It selects plain SGD at learning rate 10^{-2} and one pass; four passes differ by only 0.04 accuracy points, while the next two declared learning rates fail because they produce invalid covariance spectra. The selected configuration is frozen before five predetermined final runs. The matched no-inhibition comparison, Local SupCon grid, objective interventions, and direct-covariance control remain validation-only. No final result selects among alternative TEREL configurations.

4.4 Evaluation and uncertainty

We report raw accuracies, means, sample standard deviations, and small-sample Student-t intervals. Contrasts are paired only when methods share the same runs; comparisons across independently seeded evaluation sets are descriptive. Effective rank, feature variance, off-diagonal correlation, nearest-centroid accuracy, and between/within-class scatter diagnose representation geometry. The supplement contains the frozen protocol, selection ledger, and raw output for every reported run.

5 Results

5.1 Samplewise TeReL learns a useful representation

The frozen TeReL configuration attains $95.84 \pm 0.07\%$ on held-out MNIST. It processes one observation per optimizer step, sees the fitting split twice, and retains no temporal graph or optimizer state. All final runs update both feedforward and residual-lateral matrices and remain noncollapsed. Its mean accuracy is 1.46 points below the independently seeded TeReL-Offline reference.

The validation comparison changes only residual inhibition. Accuracy is $95.91 \pm 0.11\%$ with inhibition and $94.49 \pm 0.15\%$ for the no-inhibition reference. The paired gain is 1.42 points, with 95% Student-t interval $[1.32, 1.52]$. Effective rank is 147.0 for TeReL and 140.9 for the reference; the accuracy gain does not come from a lower-dimensional collapse.

5.2 What the residual state changes

In the final runs, the inhibitory pass reduces the root-mean-square neuron state from 0.585 to 0.495 in the first layer and from 0.675 to 0.490 in the second. The effect is systematic across runs and is largest in the deeper layer. The residual state is therefore an active part of the learned dynamics, not a relabeling of the exact gradient.

Figure 2 shows the state itself on a chronological CAPTURE-24 segment. The four randomly selected neurons share the same input interval but respond with different signs, amplitudes, and marginal distributions. This is the quantity paired with presynaptic activity in the feedforward update and with another neuron state in the lateral update.

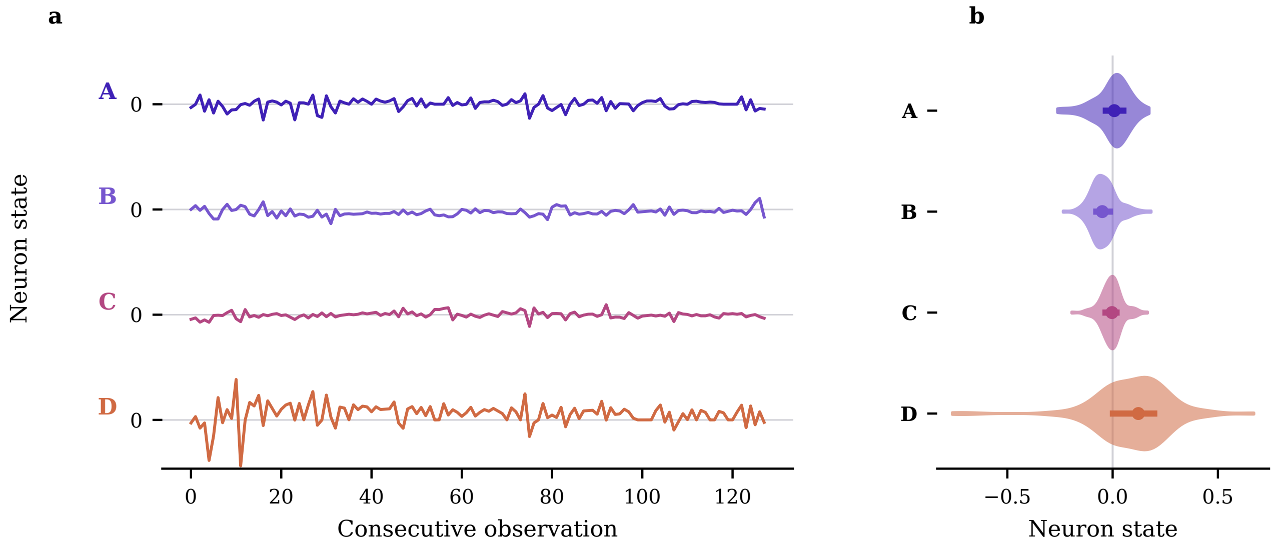

Figure 2: Neuron state on a natural sensor stream. **a**, State of four second-layer neurons during the same 128 consecutive CAPTURE-24 observations. The neurons, denoted A–D, were selected uniformly at random before the traces were inspected. **b**, Distributions over the identical interval; dots mark medians and bars mark interquartile ranges.

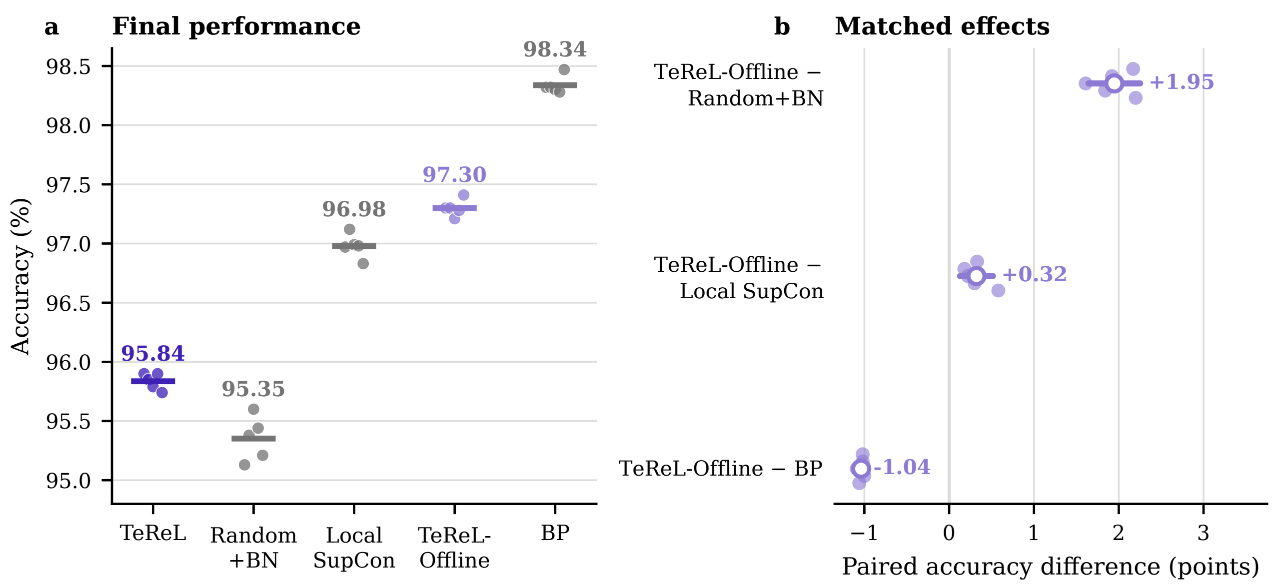

Figure 3: MNIST accuracy under the controlled temporal relation. **a**, Final run values and means for the samplewise method and the principal batched references. Independently seeded protocols are shown for context, not paired inference. **b**, Paired differences for methods evaluated on common runs under the batched protocol. Open circles mark means; lines show 95% Student-t intervals.

5.3 The batched relaxation provides a performance reference

TeReL-Offline attains $97.30 \pm 0.07\%$ test accuracy. The matched backpropagation encoder attains $98.34 \pm 0.08\%$; their paired gap is 1.04 points, with interval $[0.99, 1.08]$. The BatchNorm-calibrated random encoder attains $95.35 \pm 0.19\%$, and TEREL-Offline improves on it by 1.95 points with interval $[1.64, 2.25]$. The unnormalized random encoder attains $95.13 \pm 0.27\%$.

Local SupCon attains $96.98 \pm 0.10\%$ under the same batched protocol. TEREL-Offline is higher by 0.32 points, with paired interval $[0.13, 0.52]$. Local SupCon uses every same-label positive in its minibatch, whereas TEREL-Offline sees only adjacent examples within each constructed chunk. The small benchmark-specific difference does not establish general superiority over local contrastive learning.

5.4 The three objective terms have distinct roles

The validation-only mechanism study changes one factor at a time under the TEREL-Offline protocol. Removing the temporal term reduces accuracy by 8.18 points without collapsing effective rank. Shuffling the class-persistent order reduces accuracy by 7.55 points. Temporal coherence supplies the task-aligned teaching signal in this controlled experiment.

The other two terms prevent different degeneracies. Without gated variance expansion, the median feature variance falls to 7.7×10^{-5}. Without decorrelation, feature variance remains high but effective rank falls to 2.8. BatchNorm alone prevents neither failure. Together, these interventions assign distinct empirical functions to the temporal, variance, and decorrelation terms, as predicted by the soft-SFA formulation.

The representation operator A is also an approximation rather than a hidden same-batch gradient. Its preceding-minibatch direction has mean cosine alignment 0.76 with the same-batch direction in both layers. Relative ℓ_2 error is 0.65, and its norm is about 0.78 times the target norm. The stored direction is informative but not equivalent to the same-batch gradient.

Replacing the proxy by the exact differentiable same-batch penalty and matching the factor-four gradient scale gives $96.93 \pm 0.07\%$ validation accuracy, compared with $97.19 \pm 0.09\%$ for the measured lagged configuration. The paired direct-minus-lagged difference is -0.25 points, with 95% Student-t interval $[-0.42, -0.08]$. The measured discrepancy rules out identifying the lagged state with the exact gradient, while the accuracy comparison shows that the result does not require the exact same-batch operator.

6 Limitations

The evidence uses a constructed, label-derived temporal relation. It shows that the rule can exploit informative adjacency; it does not show that useful adjacency can be discovered without labels. Rapid or aliased latent causes can make neighboring observations misleading. Video, audio, and other continuous streams require direct evaluation across several independent sequences and timescales.

Local in time does not mean small in width. The two dense same-layer operators require $O(D^2)$ auxiliary parameters and communication; their use and update add two matrix–vector products and two rank-one products per observation. Sparse or low-rank operators are promising only if they preserve the representation gain. The implementation also uses an exact activation derivative and synchronized dense linear algebra. The locality claim concerns information paths; it is not a claim of biological realism, low energy, or efficient present-day hardware.

Finally, TEREL and TEREL-Offline differ in more than temporal detachment. They use different batch sizes, normalization, schedules, and numbers of optimizer updates. Their accuracy gap cannot be assigned to any one of these factors without a matched optimization study. The matched sample-wise comparison isolates residual-state inhibition; TEREL-Offline indicates the performance available after relaxing the execution constraints.

7 Conclusion

The soft-SFA objective defines more than three gradient terms: it defines a regularized target. Tracking the residual to that target gives each neuron a state that factors the feedforward gradient and supplies both endpoints of an anti-Hebbian lateral update. TEREL uses this construction without an error path across layers or time.

Under controlled MNIST adjacency, residual-state inhibition improves a matched samplewise reference by 1.42 points and the frozen method reaches $95.84 \pm 0.07\%$ in final evaluation. The batched relaxation reaches 97.30%, while mechanism controls distinguish temporal teaching, variance support, and decorrelation in that relaxation. These results establish deep slow-feature learning local in space and time for the controlled relation. Whether natural temporal order supplies a useful learning signal remains open.

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A Extended comparisons and diagnostics

Table 1: Comparison with the nearest objective and execution precedents. Auxiliary costs exclude learned feedforward parameters.

Method	Temporal objective	Execution	Learned mapping	Extra storage
Batch SFA	derivative minimization	batch eigensystem	linear	covariance and whitening
IncSFA	derivative minimization	sample-based	linear; deep cascades possible	incremental PCA and MCA state
Bio-SFA	derivative minimization	online neural dynamics	linear network in reported derivation	synaptic/dual variables
Greedy InfoMax	natural-order InfoNCE	isolated modules	deep nonlinear modules	context/predictor and negatives
CLAPP	causal contrastive prediction	local in depth and time	deep nonlinear layers	predictive input, modulation, negatives
LoCo	paired-view contrastive	overlapping minibatch blocks	deep nonlinear blocks	projection heads and negatives
Blockwise SSL	paired-view invariance	minibatch blocks	deep nonlinear blocks	batch statistics
TeReL	temporal soft-SFA	one observation	deep nonlinear layers	$3D$ causal scalars, one flag; $2D^2$ auxiliary parameters

Table 2: Samplewise TeReL evidence. Final accuracy is mean $\pm$ sample standard deviation. The inhibition effect is paired on the validation split; brackets give a 95% Student-t interval.

Evidence role	Accuracy / effect (points)	Effective rank
Final evaluation	95.84 ± 0.07	142.73 ± 2.09
Inhibition effect	1.42 [1.32, 1.52]	$140.94 \rightarrow 147.00$

Table 3: MNIST accuracy for the batched reference protocol. Every method uses the same concatenated-layer readout, except the explicitly marked final-layer row. Values report mean ± sample standard deviation.

Method	Accuracy
TEREL-Offline (all layers)	97.30 ± 0.07
TEREL-Offline (last layer)	97.14 ± 0.10
Random, no normalization (all layers)	95.13 ± 0.27
Backpropagation (all layers)	98.34 ± 0.08

Table 4: Paired differences in MNIST test accuracy. Values are percentage points with 95% Student-t intervals over five seed-paired differences.

Contrast	Difference [95% interval]
TEREL-Offline − BP	-1.04 [-1.08, -0.99]
TEREL-Offline − random, no normalization	2.17 [1.83, 2.51]
last − all layers	-0.16 [-0.37, 0.05]

Table 5: MNIST accuracy of the normalization-matched random control across five paired seeds. Hidden weights and affine BatchNorm parameters remain at initialization; only running statistics are calibrated on the training split. Method rows report mean ± sample standard deviation, and contrasts report 95% Student-t intervals.

Method or contrast	Accuracy / difference
Random, no normalization	95.13 ± 0.27
BatchNorm-calibrated random	95.35 ± 0.19
TEREL-Offline	97.30 ± 0.07
TEREL-Offline − calibrated random	1.95 [1.64, 2.25]
calibrated − no-normalization random	0.22 [-0.08, 0.52]

Table 6: MNIST test accuracy for the matched label-aware comparison across five seeds. Local SupCon uses all same-label positives in each minibatch. Both methods stop credit at layer boundaries and use the same architecture, 60 data presentations, and concatenated-layer probe.

Method	Accuracy
TEREL-Offline	97.30 ± 0.07
Local SupCon	96.98 ± 0.10
TEREL-Offline − Local SupCon	0.32 [0.13, 0.52]

The bracketed value is a paired 95% Student-t interval.

Table 7: One-factor mechanism interventions under the TEREL-Offline validation protocol. Δ is the accuracy difference from the full method in percentage points.

Intervention	Accuracy (%)	Δ	r_{eff}	Median var.
Full TEREL-Offline	97.28 ± 0.12	–	39.2	0.185
$\lambda_S = 0$	89.10 ± 1.57	-8.18	369.3	0.907
$\lambda_V = 0$	66.13 ± 6.32	-31.15	6.0	7.72e-05
$\lambda_C = 0$	86.44 ± 0.41	-10.84	2.8	0.916
Shuffled order	89.73 ± 0.37	-7.55	311.3	0.171

The final-layer probe of TEREL-Offline attains $97.14 \pm 0.10\%$, 0.16 points below the matched all-layer probe; the paired interval $[-0.37, 0.05]$ includes zero. The final layer therefore retains nearly all of the mean result, though five runs do not resolve the small readout difference.

Table 8: Geometry of the MNIST representations across final runs. B/W is the between/within-class scatter ratio; Sel. is median unit class selectivity; r_{eff} is covariance effective rank; and $|\rho_{\text{off}}|$ is mean absolute off-diagonal correlation.

Method	Centroid (%)	B/W	Sel.	r_{eff}	$\lvert\rho_{\text{off}}\rvert$
Random, no norm. (all)	80.21	0.241	1.239	33.84	0.101
TEREL (all)	90.58	0.082	0.833	142.73	0.058
TEREL-Offline (all)	95.55	0.695	1.704	37.35	0.103
TEREL-Offline (last)	96.22	1.516	2.567	18.86	0.161
BP (all)	97.87	0.968	1.857	21.17	0.152

Table 9: Alignment of the lagged lateral direction with its same-batch target on validation data. Statistics average three runs and all nonzero minibatches.

Layer	Cosine alignment	Relative ℓ_2 error	Norm ratio
1	0.757	0.651	0.785
2	0.755	0.653	0.780

B Gradient derivations

For detached p, μ, σ^2, A, the three single-observation activation derivatives are

$$\frac{\partial \ell_S}{\partial z_t} = \frac{2I_t}{D}(z_t - p_t), \tag{15}$$

$$\frac{\partial \ell_V}{\partial z_t} = -\frac{2}{D} g \odot (z_t - \mu), \tag{16}$$

$$\frac{\partial \ell_C}{\partial z_t} = \frac{1}{D} A(z_t - \mu). \tag{17}$$

Factoring out $2\lambda_S/D$ leaves exactly r_t in Eq. (3). Because $\hat{z}_t$ is detached,

$$\frac{\partial}{\partial z_t}\frac{\lambda_S}{D}\|z_t - \hat{z}_t\|^2 = \frac{2\lambda_S}{D} r_t. \tag{18}$$

Mapping through the activation gives $b_t = J_{\phi,t}^\top r_t$. At zero residual-state coupling, $s_t = b_t$ and the preactivation target obeys

$$\frac{\partial}{\partial W}\frac{\lambda_S}{D}\|a_t - \mathrm{sg}(a_t - b_t)\|^2 = \frac{2\lambda_S}{D} b_t x_t^\top, \tag{19}$$

which equals the chain rule for Eq. (2). Nonzero κ replaces b_t by the inhibited state s_t and deliberately modifies this direction. With an undetached temporal reference, z_t also appears as the predecessor in pair $t+1$, yielding Eq. (14).

TEREL-Offline uses $h_{\mathcal{B}} = N_{\theta_N,\mathcal{B}}(a_{\mathcal{B}})$ and $z_t = \phi(h_t)$, where N is training-mode BatchNorm. Let $e_s = \partial L/\partial z_s$, $r_s = e_s \odot \phi'(h_s)$, and $J_{st}^N = \partial h_s/\partial a_t$. Its exact same-layer chain rule is

$$u_t = \sum_{s\in\mathcal{B}} (J_{st}^N)^\top r_s, \qquad \frac{\partial L}{\partial W} = \sum_t u_t x_t^\top, \qquad \frac{\partial L}{\partial c} = \sum_t u_t, \tag{20}$$

with $\partial L/\partial\theta_N = \sum_s (\partial h_s/\partial\theta_N)^\top r_s$ for the affine parameters. BatchNorm mixes examples within each coordinate; this expression admits no error from another learned layer.

For the direct covariance control, let Z_c stack centered rows and $C = \mathrm{offdiag}(Z_c^\top Z_c/B)$. Since C is symmetric,

$$\frac{\partial}{\partial Z_c}\frac{\|C\|_F^2}{D} = \frac{4}{BD} Z_c C, \tag{21}$$

which is $4C(z_t - \mu)/(BD)$ in column-vector notation. TEREL replaces C by the stored A and absorbs the positive factor into λ_C.

C Implementation-fidelity tests

The public test suite verifies the following properties:

- equality between the soft-objective gradient and the detached target gradient at zero residual-state coupling, including the variance sign and temporal boundary;
- the presynaptic–neuron-state outer product, one-pass inhibitory correction, and use of the same detached states in the lateral moment;
- nonzero updates to every encoder layer and both lateral states, absence of cross-layer gradients, fixed causal-state size, and release of every samplewise graph;
- training-split-only BatchNorm calibration that changes running buffers while preserving every random-encoder weight, bias, and affine parameter;
- the MNIST split roles, normalization fit only on the training split, temporal boundaries, and encoder batches containing only features and boundary indicators;
- matched all-layer and final-layer probes, greedy layerwise scheduling, execution of every declared backpropagation layer, and exact counting of data presentations;
- batch-SFA constraints, boundary-aware derivatives, and numerical agreement of the published-algorithm IncSFA port with its reference; and
- refusal to execute final evaluation after any change to the frozen protocol, selection ledger, configuration, or tracked source.

D Reproducibility record

The repository fixes Python 3.12 and all direct dependencies through `pyproject.toml` and `uv.lock`. Every run enables deterministic PyTorch and cuDNN behavior together with the prescribed cuBLAS workspace configuration. The anonymous supplement contains the selected configurations, selection ledger, seeds, per-seed JSON records, raw confusion matrices, hardware and software record, source, generated tables, and checksums. `ARTIFACT_README.md` provides portable commands and the exact provenance identifiers. The supplement does not redistribute dataset files.

E Complete configurations and raw results

Table 10: Raw final-run values for samplewise TEREL. Accuracy is in percent.

Field	Values
Seeds	42, 43, 44, 45, 46
Accuracy	95.90, 95.85, 95.79, 95.90, 95.74
Effective rank	140.71, 144.54, 144.91, 140.45, 143.03

Table 11: Training resources for samplewise TEREL. Causal state, auxiliary parameters, feedforward parameters, and optimizer state are reported separately.

Updates	Causal values	Auxiliary parameters	Feedforward parameters	Optimizer bytes
100,000	2,306	655,360	533,248	0

Table 12: MNIST test accuracy for seeds 1101, 1202, 1303, 1404, and 1505, in that order.

Method	Raw values
TEREL-Offline (all layers)	97.30, 97.30, 97.21, 97.28, 97.41
TEREL-Offline (last layer)	97.14, 97.18, 97.28, 97.09, 97.01
Random, no normalization (all layers)	94.66, 95.27, 95.12, 95.33, 95.27
Backpropagation (all layers)	98.32, 98.32, 98.30, 98.28, 98.47

Table 13: Resource use during encoder fitting, averaged across five seeds. Examples count sample presentations. MAC denotes the declared linear and same-layer operation proxy, not energy use.

Method	Examples (M)	Opt. steps	Seconds	Peak MiB	MAC (G)
TEREL-Offline (all layers)	3.00	11760	107.6	76.7	3981.3
TEREL-Offline (last layer)	3.00	11760	107.7	76.7	3981.3
Random, no normalization (all layers)	0.00	0	0.0	0.0	0.0
Backpropagation (all layers)	3.00	11760	43.4	225.4	4815.4

Table 14: Training storage in MiB. Model parameters include every parameter fitted by the optimizer, including the supervised head for backpropagation. Auxiliary storage contains TeReL-Offline lateral and statistical buffers and excludes normalization buffers. Optimizer and auxiliary storage are training-only. The inference encoder retains hidden-layer parameters and normalization buffers but discards the supervised head and auxiliary buffers.

Method	Model params	Optimizer	Auxiliary	Norm.	Inference encoder
TeReL-Offline (all layers)	2.040	4.080	1.262	0.006	2.046
TeReL-Offline (last layer)	2.040	4.080	1.262	0.006	2.046
Random, no normalization (all layers)	2.034	0.000	0.000	0.000	2.034
Backpropagation (all layers)	2.050	4.100	0.000	0.006	2.046

Table 15: Local SupCon selection and final record. The left block reports the validation grid; bold marks the configuration selected before final evaluation. The right block reports the resource state of that configuration.

Validation selection

Learning rate	Temperature	Accuracy
1e-04	0.1	96.49 ± 0.15
1e-04	0.2	95.60 ± 0.12
3e-04	0.1	96.63 ± 0.03
3e-04	0.2	95.98 ± 0.11
1e-03	0.1	**96.90 ± 0.03**
1e-03	0.2	96.20 ± 0.02

Final record

Quantity	Value
Mean accuracy (%)	96.98 ± 0.10
Mean encoder seconds	82.1
Model parameters (MiB)	2.040
Optimizer state (MiB)	4.080
Normalization buffers (MiB)	0.006
Inference encoder (MiB)	2.046

Final accuracies (%): 96.97, 97.12, 96.99, 96.98, 96.83.